NicheTarget-GNN: niche-aware in-silico target ranking, tested across 17 patients

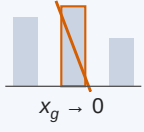
Spatial cell graph

cells + LR edges

Niche-aware GNN

LR-edge attention + niche conditioning

In-silico inhibition



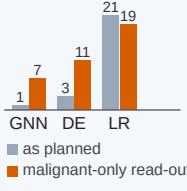
predict read-out shift per niche

Niche-specific ranked targets

1 MLPH
2 ANKRD30A
3 CCND1
4 KRT7
5 GATA5

Xenium breast Rep1 (GNN ensemble, top 5)

Cross-patient replication



top-25 genes shared with discovery tumour